



US012396441B2

(12) **United States Patent**
Fries et al.

(10) **Patent No.:** **US 12,396,441 B2**

(45) **Date of Patent:** **Aug. 26, 2025**

(54) **BIOMIMETIC SENTINEL REEF
STRUCTURES FOR OPTICAL SENSING AND
COMMUNICATIONS**

(71) Applicants: **David Fries**, Pensacola, FL (US); **Tim
Hutcheson**, Pensacola, FL (US); **Noam
Josef**, Pensacola, FL (US); **David
Millie**, Pensacola, FL (US); **Connor
Tate**, Pensacola, FL (US)

(72) Inventors: **David Fries**, Pensacola, FL (US); **Tim
Hutcheson**, Pensacola, FL (US); **Noam
Josef**, Pensacola, FL (US); **David
Millie**, Pensacola, FL (US); **Connor
Tate**, Pensacola, FL (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/419,998**

(22) Filed: **Jan. 23, 2024**

(65) **Prior Publication Data**

US 2024/0164353 A1 May 23, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/134,759, filed on
Dec. 28, 2020, now Pat. No. 11,877,563, and a
continuation-in-part of application No. 16/273,246,
filed on Feb. 12, 2019, now abandoned.

(60) Provisional application No. 62/629,249, filed on Feb.
12, 2018.

(51) **Int. Cl.**
E02B 3/04 (2006.01)
A01K 61/73 (2017.01)
A01K 61/78 (2017.01)

(52) **U.S. Cl.**
CPC **A01K 61/73** (2017.01); **A01K 61/78**
(2017.01)

(58) **Field of Classification Search**
CPC E02B 3/129; E02B 3/046; E02B 3/043;
E02B 2201/04; E02B 17/00; Y02A 40/81;
Y02A 10/26; Y02A 40/80; Y02A 10/00;
Y02A 10/11; A01K 61/73
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

411,606 A * 9/1889 Parker A01M 23/32
43/90
433,241 A * 7/1890 Hawkins, Jr. A01M 23/32
43/63
482,635 A * 9/1892 Hanish A01K 77/00
43/12
1,336,356 A * 4/1920 Johnson A01K 69/00
43/105
1,635,899 A * 7/1927 Nohlen E02B 3/04
405/28
2,603,031 A * 7/1952 Haseman A01K 69/06
43/105
3,029,546 A * 4/1962 Ruiz A01K 77/00
43/105
3,119,407 A * 1/1964 Timmerman, Sr. F16K 21/20
141/95

(Continued)

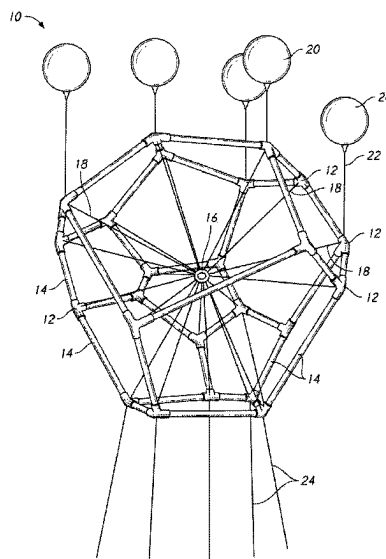
Primary Examiner — Edwin J Toledo-Duran

(74) *Attorney, Agent, or Firm* — J. Wiley Horton

(57) **ABSTRACT**

An artificial reef structure including thin tensile members as
part of the structure. These tensile members readily accu-
mulate marine life to create a more organic structure. The
reef structure can accommodate a wide variety of ocean
sensing and communication systems. Embedded lighting
and electronic elements can be used to attract and monitor
marine life.

6 Claims, 7 Drawing Sheets



US 12,396,441 B2

Page 2

(56)	References Cited		5,347,745 A *	9/1994	Authement	A01K 75/04
	U.S. PATENT DOCUMENTS					43/44.91
			5,427,472 A *	6/1995	Ono	E02B 3/06
						405/21
3,133,443 A *	5/1964	Timmerman, Sr.	5,477,798 A *	12/1995	Ness	B63B 3/13
		B65D 90/48				114/65 R
3,215,108 A *	11/1965	Otis	5,480,261 A *	1/1996	Meyers	E02B 15/0885
		B63B 35/44				405/63
		52/654.1	5,556,229 A *	9/1996	Bishop	E02B 3/04
3,347,210 A *	10/1967	Golub				405/32
		A01K 61/54	5,617,813 A *	4/1997	Loverich	A01K 61/60
		119/238				119/223
3,422,628 A *	1/1969	McDonald	5,628,279 A *	5/1997	Bones, IV	A01K 61/60
		E02B 17/00				119/223
		137/236.1	5,713,303 A *	2/1998	Willinsky	A01K 61/17
3,541,721 A *	11/1970	Sexauer				119/223
		A01K 69/10	5,797,348 A *	8/1998	Taherzadeh	A01K 61/80
		43/100				119/223
3,628,333 A *	12/1971	Newton	6,035,575 A *	3/2000	Hilty	A01K 69/06
		E02B 15/06				43/100
		210/242.3	6,106,194 A *	8/2000	Angel	B66C 1/12
3,698,359 A *	10/1972	Fremont				405/17
		A01K 61/60	6,186,702 B1 *	2/2001	Bartkowski	A01K 61/70
		119/233				119/221
3,701,261 A *	10/1972	Nolan, Jr.	6,234,715 B1 *	5/2001	Ono	E02B 3/06
		E21B 43/013				405/21
		405/188	6,318,931 B1 *	11/2001	Khachaturian	B63B 27/10
3,803,855 A *	4/1974	Malkiel				405/209
		B65D 88/78	6,327,994 B1 *	12/2001	Labrador	B63B 39/06
		114/257				114/382
3,830,066 A *	8/1974	Larsen	6,367,399 B1 *	4/2002	Khachaturian	B63B 77/00
		E02B 3/04				114/61.1
		405/25	6,931,812 B1 *	8/2005	Lipscomb	E04B 1/19
3,864,049 A *	2/1975	Ono				52/653.1
		E02B 3/06	7,448,827 B2 *	11/2008	Li	E02D 29/0216
		403/171				405/116
3,879,951 A *	4/1975	Mason	7,540,814 B2 *	6/2009	Dilling	A63B 63/00
		E21B 43/0122				473/476
		405/60	7,612,735 B2 *	11/2009	Essig, Jr.	A01K 61/60
3,898,958 A *	8/1975	Pranis, Jr.				343/915
		E02B 3/046	D607,081 S *	12/2009	Harper	D22/199
		119/221	D607,082 S *	12/2009	Harper	D22/199
3,903,637 A *	9/1975	Dorsey	7,681,528 B2 *	3/2010	Ytterland	A01K 61/60
		A01K 69/10				119/203
		43/105	7,686,543 B2 *	3/2010	They	E01D 19/02
3,922,868 A *	12/1975	McDonald				405/224.1
		E02B 17/027	7,748,349 B2 *	7/2010	Thorvardarson	A01K 61/60
		114/265				119/223
4,058,945 A *	11/1977	Knapp	8,683,955 B2 *	4/2014	Madsen	A01K 61/65
		E04B 1/32				119/223
		D25/4	2002/0185079 A1 *	12/2002	Truebe	E02B 1/006
4,074,497 A *	2/1978	Ono				119/219
		E02B 3/06	2003/0168864 A1 *	9/2003	Heronemus	B63B 35/44
		405/35				290/55
4,290,714 A *	9/1981	Strange	2004/0148901 A1 *	8/2004	Cadogan	E04H 15/20
		E21B 41/005				52/646
		405/60	2005/0115186 A1 *	6/2005	Jensen	E01F 9/696
4,350,143 A *	9/1982	Laing				52/633
		F24S 80/56	2005/0235921 A1 *	10/2005	Niezrecki	A01K 61/60
		126/567				119/223
4,373,286 A *	2/1983	Robison	2006/0033674 A1 *	2/2006	Essig	F24S 23/715
		A01K 69/00				343/915
		43/10	2006/0056914 A1 *	3/2006	Arai	E02B 3/046
4,406,151 A *	9/1983	Simonsen				405/30
		E02D 29/073	2006/0062676 A1 *	3/2006	Jakubowski	B63B 35/44
		72/481.2				416/244 R
4,409,752 A *	10/1983	Anderson	2007/0039254 A1 *	2/2007	Onda	B63C 11/49
		A01K 69/06				52/81.1
		43/105	2008/0110408 A1 *	5/2008	Thorvardarson	A01K 61/10
4,437,794 A *	3/1984	Grimsley				119/223
		E02B 17/00	2009/0035068 A1 *	2/2009	Teraï	B63G 9/04
		405/203				405/211
4,447,978 A *	5/1984	Robison	2010/0018470 A1 *	1/2010	Kim	A01K 61/60
		A01K 69/00				119/223
		43/10	2010/0229545 A1 *	9/2010	Griffin	F03D 9/008
4,462,535 A *	7/1984	Johnston				60/497
		E02D 29/063				
		29/469				
4,471,552 A *	9/1984	McIntosh				
		A01K 79/02				
		43/4.5				
4,492,270 A *	1/1985	Horton				
		E02B 17/021				
		405/196				
4,509,285 A *	4/1985	Smith				
		A01K 73/00				
		43/17.1				
4,645,379 A *	2/1987	Grimsley				
		B63B 35/44				
		D25/1				
4,679,961 A *	7/1987	Stewart				
		E04B 1/19				
		24/654				
4,957,064 A *	9/1990	Koma				
		A01K 61/60				
		119/223				
5,027,735 A *	7/1991	Labrador				
		B63B 43/06				
		114/39.29				
5,044,828 A *	9/1991	Berner, Jr.				
		E02B 17/0004				
		52/651.07				
5,056,447 A *	10/1991	Labrador				
		B63H 9/069				
		416/8				
5,172,649 A *	12/1992	Bourgeois				
		A01K 61/60				
		119/223				
5,265,395 A *	11/1993	Lalvani				
		E04B 1/32				
		403/176				

(56)	References Cited		2015/0063910 A1 *	3/2015	Meltsov	E02B 8/00 114/230.14
	U.S. PATENT DOCUMENTS		2015/0136012 A1 *	5/2015	Williams	B25J 9/08 901/1
2011/0031062 A1 *	2/2011	Elmer	2015/0150223 A1 *	6/2015	Robinson	A01K 61/60 119/223
2011/0174232 A1 *	7/2011	Hoie	2015/0152614 A1 *	6/2015	Burt	B28B 7/00 405/23
2011/0296908 A1 *	12/2011	Kjerstad	2015/0167288 A1 *	6/2015	Harkin	B64D 11/064 403/107
2011/0318106 A1 *	12/2011	Gateff	2015/0330046 A1 *	11/2015	Betcher	E02B 3/062 405/26
2012/0006277 A1 *	1/2012	Troy	2016/0183501 A1 *	6/2016	Page	A01K 61/60 119/223
2012/0132123 A1 *	5/2012	Khachaturian	2017/0036782 A1 *	2/2017	Dula	B64G 1/402
2012/0167829 A1 *	7/2012	Madsen	2017/0055502 A1 *	3/2017	Gagliano	E02B 3/046
2012/0199078 A1 *	8/2012	Krone	2017/0233965 A1 *	8/2017	Boasso	E02B 3/121 405/20
2013/0129421 A1 *	5/2013	Belzile	2018/0080744 A1 *	3/2018	Osienski	F41H 11/05
2013/0320678 A1 *	12/2013	Kittel	2018/0317461 A1 *	11/2018	Meeker	A01K 61/10
2014/0138955 A1 *	5/2014	Fransen	2019/0029231 A1 *	1/2019	Becher	A01K 61/60
2014/0158171 A1 *	6/2014	Doleshal	2019/0150410 A1 *	5/2019	Dwyer	A01K 61/80
2014/0356069 A1 *	12/2014	Gateff	2019/0340440 A1 *	11/2019	Atwater	G06T 7/20
2015/0023734 A1 *	1/2015	Higgins	2019/0387721 A1 *	12/2019	Dobson	A01K 61/60
			2020/0281168 A1 *	9/2020	Hellesøe	H02S 10/40
			2020/0281169 A1 *	9/2020	Hellesøe	A01K 61/80
			2021/0400927 A1 *	12/2021	Østerhus	F03B 13/142
			2022/0132819 A1 *	5/2022	Chadwick	A01K 69/08 43/102
			2022/0183261 A1 *	6/2022	Gray	A01K 61/55
			2024/0401300 A1 *	12/2024	Yang	E02B 3/16

* cited by examiner

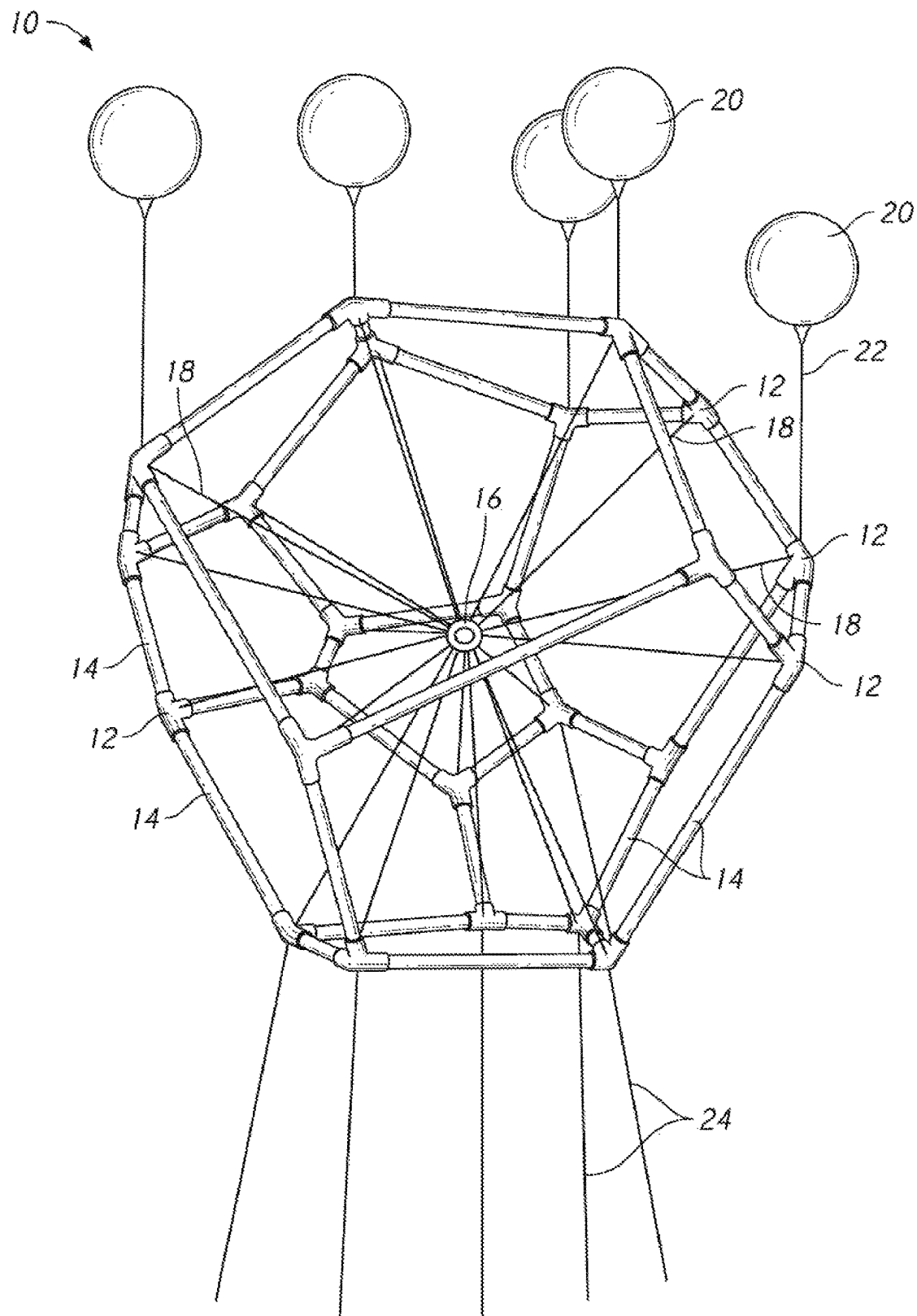


FIG. 1

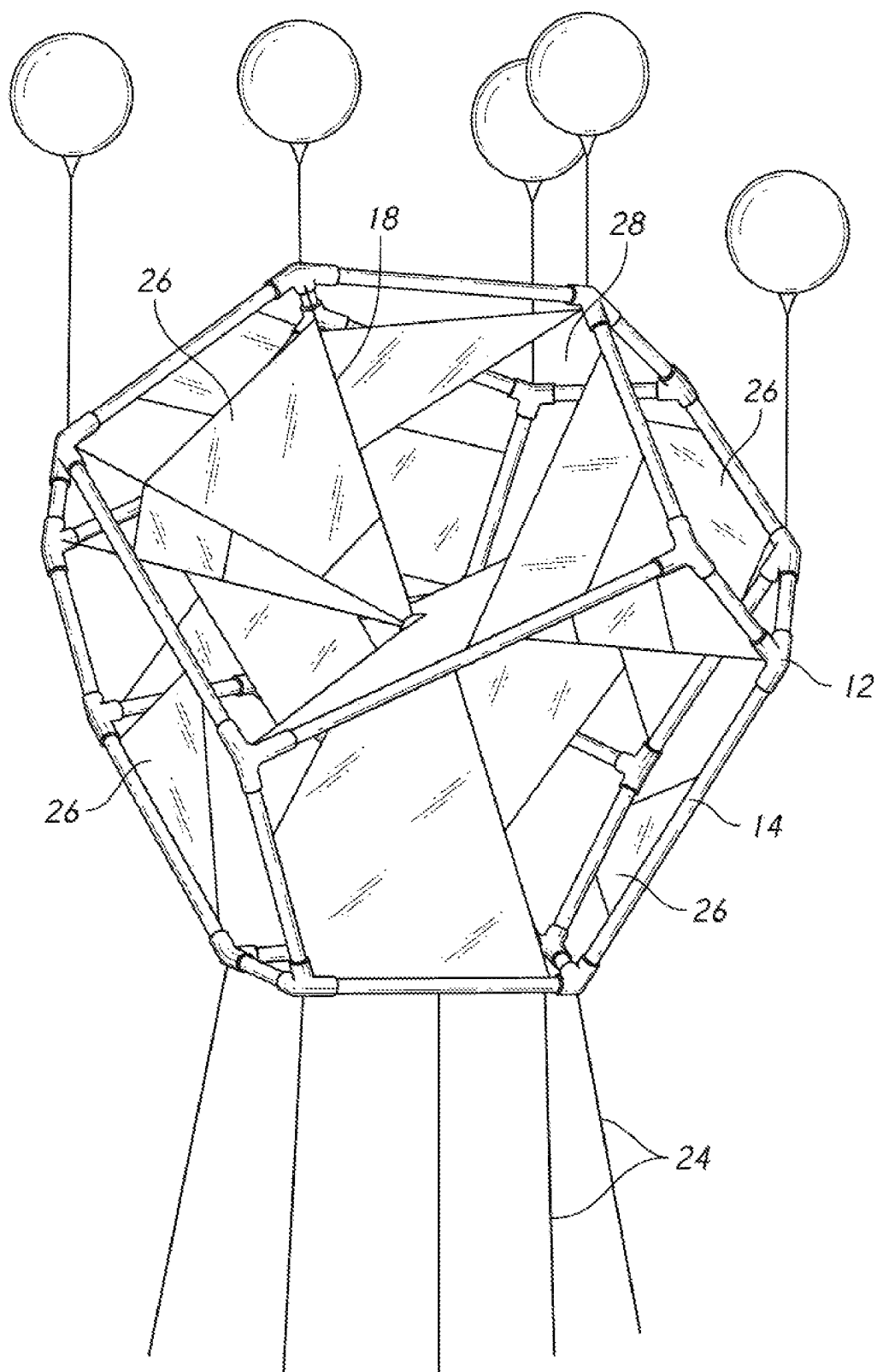


FIG. 2

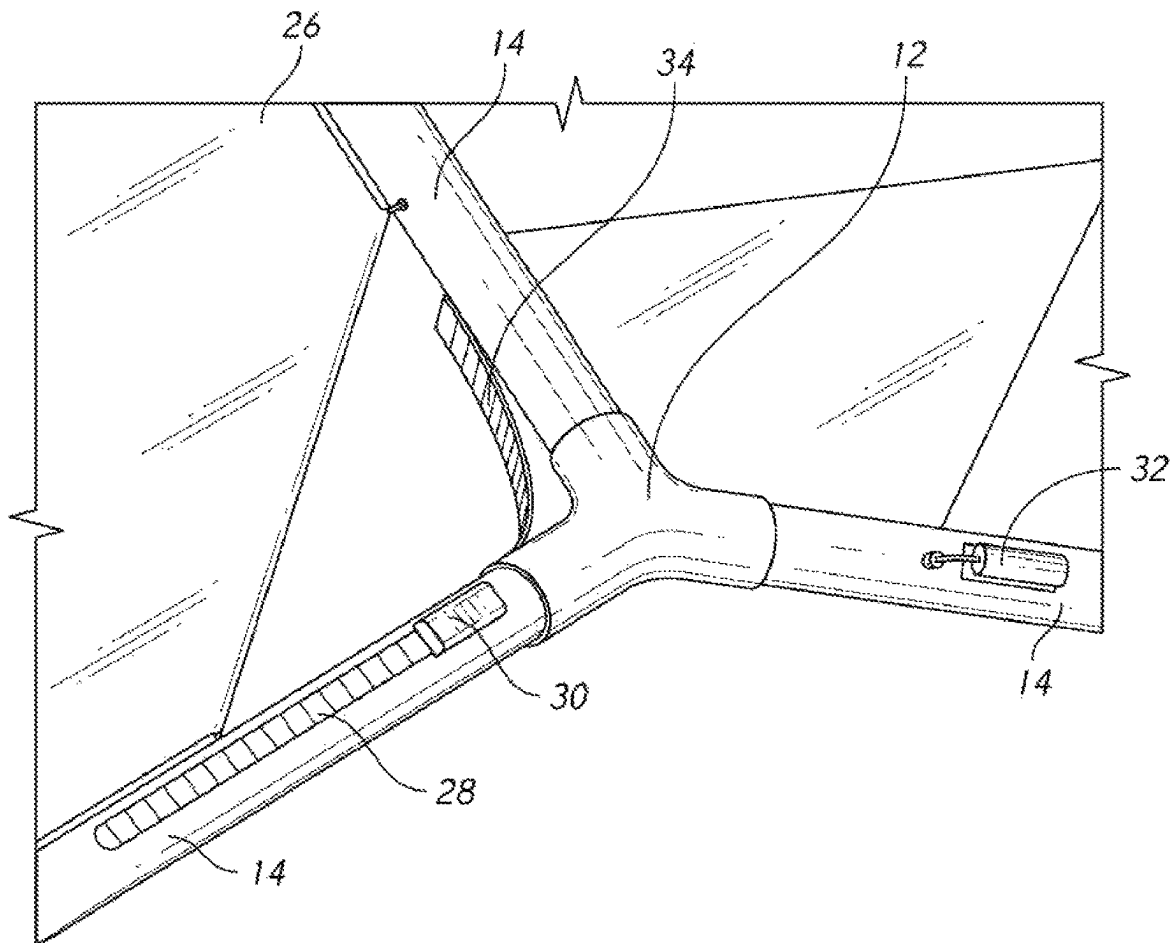


FIG. 3

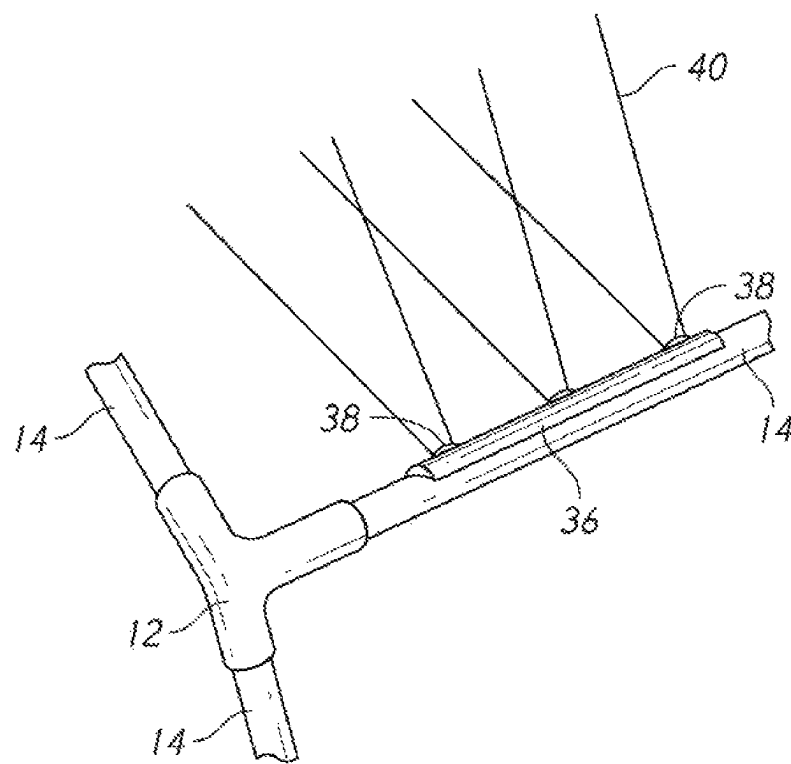


FIG. 4

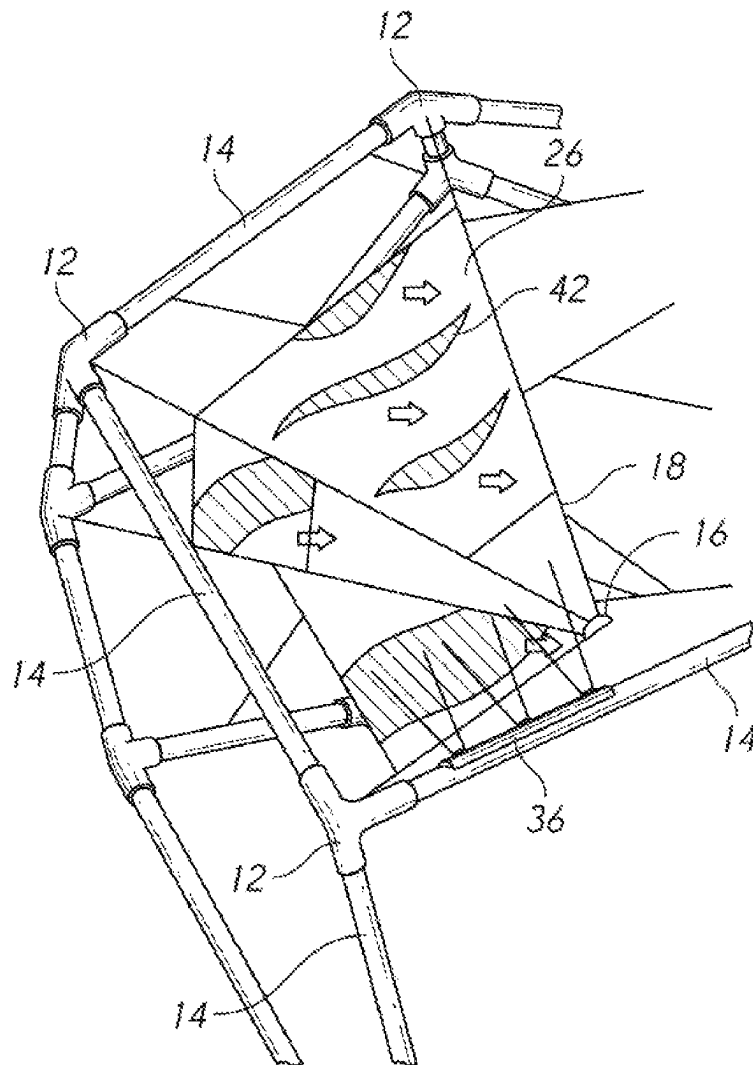


FIG. 5

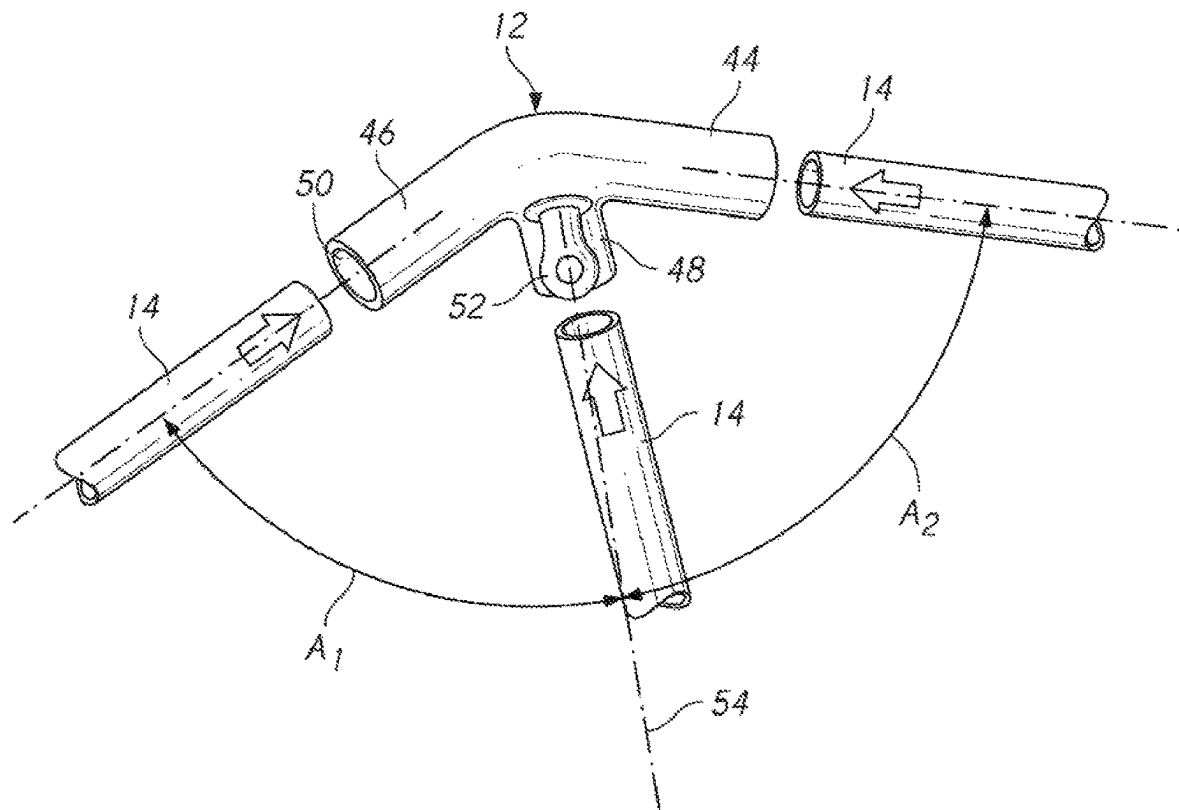


FIG. 6

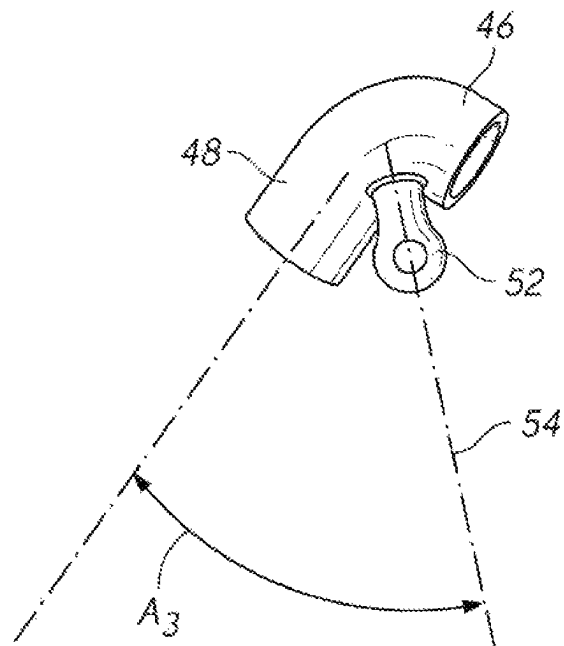


FIG. 7

1

BIOMIMETIC SENTINEL REEF STRUCTURES FOR OPTICAL SENSING AND COMMUNICATIONS

CROSS-REFERENCES TO RELATED APPLICATIONS

This non-provisional patent application is a continuation of U.S. patent application Ser. No. 17/134,759 (now U.S. Pat. No. 11,877,563). The parent application listed the same inventors.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not applicable

MICROFICHE APPENDIX

Not Applicable

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention pertains to the field of artificial reefs. More specifically, the invention comprises a novel reef structure that incorporates optical transmission, sensing, and communications.

2. Description of the Related Art

Many current beaches have little fish habitat. Where no vertical structure is present, fish have nowhere to spawn and hide from predators. Fisheries refugia (natural and artificial reefs) are marine areas in which specific management measures are applied to sustain important species. Natural fisheries refugia can be a model for the creation of artificial reefs. Evidence from existing marine reserves indicates that the presence of reef structures increases fish abundance and diversity. For this reason, artificial reefs have been introduced in areas lacking sufficient structure.

Traditional artificial reef structures are designed with uniform cellular architectures. Their topologies do not mimic natural reef forms. A good example is an artificial reef structure made of steel reinforced concrete. These structures often include a central volume that is accessed through a series of holes in its exterior walls. Stones or shells may be added to the exterior surface to better simulate a natural reef. Even so, cast concrete structures do not tend to accurately mimic a natural reef.

On the other hand, strings and ropes are a proven fisheries and mariculture construction element. They are used throughout the world as artificial reef scaffolding. A wide variety of marine life adheres to the strings and ropes and transforms the assembly into a more organic structure. The present invention takes advantage of this fact, among other things.

BRIEF SUMMARY OF THE INVENTION

The present invention comprises an artificial reef structure including thin tensile members as part of the structure. These tensile members readily accumulate marine life to create a more organic structure. The reef structure can accommodate a wide variety of ocean sensing and communication sys-

2

tems. Embedded lighting and electronic elements can be used to attract and monitor marine life.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is perspective view, depicting an artificial reef made according to the present invention.

FIG. 2 is a perspective view, depicting the reef of FIG. 1 with the addition of a plurality of panels.

FIG. 3 is a detailed perspective view, showing the inclusion of light elements and sensory elements on the reef structure of FIG. 1.

FIG. 4 is a detailed perspective view, showing an exemplary location for a projector array.

FIG. 5 is a detailed perspective view, showing the formation of moving projected images on the sheets.

FIG. 6 is an exploded perspective view, showing the nature of the joints used in the inventive structure.

FIG. 7 is an elevation view, showing the joint of FIG. 6 from a different vantage point.

REFERENCE NUMERALS IN THE DRAWINGS

10 artificial reef
12 joint
14 link
16 connector
18 tension member
20 float
22 float line
24 mooring line
26 sheet
28 gap
29 light array
30 controller
32 instrument package
34 flexible array
36 projector array
38 projector
40 projector field of view
42 image
44 first leg
46 second leg
48 third leg
50 receiver
52 anchor point

DETAILED DESCRIPTION OF THE INVENTION

The present invention may be realized using a wide variety of structures. A single exemplary structure is depicted in FIGS. 1-3. This exemplary structure will be described in detail, but the reader should keep in mind that many other structures are possible.

The inclusion of rope or string-like tensile members is an important feature of the present invention. The structure depicted in FIG. 1 includes a great many tensile members. The structure is created by connecting links 14 together using joints 12. In this version, each joint includes three hollow cylindrical receivers configured to receive the cylindrical end of a link 14. The structure is designed so that the tension members 18 are placed in tension and links 14 are placed in compression. This allows the structure to be held together without adhesives.

3

Connector **16**—which may be an assembly of multiple pieces—is placed approximately in the open center of the structure. A tension member **18** is then connected between each joint **12** and connector **16**. Turnbuckles or similar devices may be included in each tension member **18** to gradually increase the tension until the structure as a whole is in a rigid and balanced state. Additional tension members may be added that link two existing tension members together. In other words, “cross-lacing” tension members may be added to increase the density of tension members.

The reef structure may be placed on the bottom of a body of water or it may be free-floated some distance above the bottom. Buoyancy is required for a free-floating structure. It is possible to provide the desired buoyancy by making the links and/or joints buoyant. External buoyancy may also be added. In the example of FIG. 1, a plurality of floats **20** are connected to the structure via float lines **22**. The structure is anchored to the sea floor via mooring lines **24**. Using this approach, the artificial reef may be positioned at a desired depth and retained there. It is often preferable to anchor the structure in the benthic zone proximate the floor of a body of water. The structure shown in FIG. 1 is well suited to this purpose.

FIG. 2 shows the same artificial reef structure following the addition of a set of sheets **26**. Sheets **26** may be made of a variety of materials. Some sheets are a piece of dense netting stretched between links, joints, and tension members. At least one side of the sheet is preferably angled toward the open center of the reef structure—with at least one portion of the sheet being secured within the interior of the structure. This adds geometric complexity and creates many internal passages. Some sheets may be made of PVC having various opacities. The sheets preferably do not continuously cover any portion of the reef structure. Rather, they leave gaps **28**.

Marine life (such as barnacles) will adhere to the tension members and the sheets. The marine life will grow, bridge, and fuse in unpredictable ways. The result will be a primarily organic structure where the supporting links, joints, and tension members are buried in growth. The supporting links, tension members, and joints may be made of a material which will eventually degrade and disintegrate, leaving a structure comprised only of marine life.

It is desirable for the reef to include light sensing elements, light producing elements, and various other sensing and communicating elements. FIG. 3 shows a first example. Light array **29** is placed on the interior-facing side of a link **14**. Controller **30** controls this light array. Light produced by the array tends to fall on one or more adjacent sheets **26**. The light array preferably includes multiple LED's that can be used to produce a variety of wavelengths. The array can be controlled to produce a rippling or shimmering effect that tends to attract fish to the structure.

Flexible array **34** bends around joint **12** as shown. It may be used to produce a second controlled lighting effect. Instrument package **32** contains instruments to monitor the water flowing through the reef structure (such as salinity, specific gravity, etc.). All these instruments may be centrally controlled and networked together.

FIG. 4 shows a second exemplary device that is used to create lighting effects. Projector array **36** contains three separate LED projectors **38** in a waterproof housing. Each projector **38** projects a diverging image extending across a conical field of view **40**. The projector array shown in FIG. 4 is positioned to create a moving light image on an adjacent sheet or sheets.

4

FIG. 5 shows the same projector array **36**—along with more of the surrounding structure. The projector array creates one or more images **42** on the adjacent sheets **26**. These images are configured to move as indicated by the arrows. The images need not be sharply focused. An image that is effective for attracting marine life is one that simulates the rippling light patterns produced by light refracting through surface wave action. It is preferable for the projecting system to have significant depth of field so that the plane of the sheet need not be perpendicular to the projection axis.

In order to facilitate a cost-effective deployment for the reef structure that can be achieved globally, the materials need to be universally applicable and accessible to the public and private sector. Construction materials can include concrete, PVC, polypropylene rope, 3D printed joints (selecting from confirmed non-toxic materials/FDA approved/high tensile strength including ABS, Nylon, TGlase/Pett and steel). Embedded optics may be added to the structure.

A standard design includes a polypropylene rope tension structure configured in a complex web running through the center of the module. These ropes will accrue growth to attract small demersal fish and arthropods to develop a feeding ground for stock fish. The rope tension members will also remain strong and durable with no significant degradation. In connection with the ropes are complex interstitial spaces of varied size created through the use of hollow, solid, clear and opaque PVC's. The variation in opacity and interior spaces will allow for implementation of sensing technology as well as provide visual protection to a variety of species. The stringed reef systems can be resident at the benthic layer or moored as floating reefs in the water column floating away from any substrate completely undisturbed. The lightweight design and flexible string scaffolding makes these materials more durable against the elements and the materials used are designed against corrosion.

The distributed cables and string approach allows three-dimension fluid flow through the structure. The three-dimension structure provides protective spaces and also allows contiguous paths for nutrients, light and signaling. Past work by others (Shashar and Josef et. al.) in the Red Sea exhibit successful reef productivity for corals and fish within a coarse mesh volume. Additional work for oysters in local Gulf of Mexico waters in the Pensacola area used wired cabled mesh system with highly successful reef development, health, and resilience.

Returning now to FIG. 1, some of the significant aspects of the inventive structure will be explained in more detail. As stated previously, the structure is configured to use tension in tension members **18** to place all links **14** in compression. The reader will note that the overall shape of the exterior is made of intersecting planar “rings.” Most of the planar rings in the configuration shown have five links **14** joined by six joints **12**. The links in each ring lie primarily in a single plane. The word “primarily” is used because the links are preferably made of flexible material. Some out-of-plane flexing is allowed to occur as the structure is drawn tight. This allowance means that the dimensions are not critical. In fact, a single ring can be twisted by as much as 30 degrees.

FIGS. 6 and 7 provide more details regarding the components comprising the structure shown in FIG. 1. Each joint **12** includes three legs configured to receive the ends of three links **14**. First leg **44**, second leg **46**, and third leg **48** all include a receiver **50** sized to slidably receive the outer diameter of a link **14**. An end of each link **14** can be slid into joint **12** as indicated by the arrows.

5

Anchor point **52** is provided on the side of joint **12** facing inward toward connector **16** (shown in FIG. 1). The angles between the three legs are set to ensure that a tension member attached to anchor point **52** will place the links in compression. Axis **52** indicates the position of a tension member attached to anchor point **52**.

The geometry of the three legs **44,46,48** is set so that the angle between the centerline of any leg and axis **54** will be less than 90 degrees (an acute angle). The reader will note that angle **A1** is an acute angle and that angle **A2** is also an acute angle. FIG. 7 shows the same joint **12** from the side. The reader will observe that the angle **A3** between the centerline of third leg **48** and axis **54** is also an acute angle.

The same is true for each of the legs in each of the joints. The angle between all the legs and the axis of the tension member coming off a particular joint will always be acute. This fact means that tension placed on the tension members tends to pull the entire structure inward and place all the links in compression.

Light has been used for centuries to attract marine organisms and light in the reef is a crucial cue for attracting larvae. Underwater light contrast and color also play important roles in the fish larvae's capability to detect a potential settlement habitat. The inventive artificial reef includes light arrays as shown in FIG. 3. These can be used to mimic natural patterns and thereby attract selected species.

The inventive reef preferably includes optical sensing as well. Underwater optical sensing and mobility can be challenging due to the absorptive properties of water for many of the optical signals necessary for navigation, communications and control. Optical sensing beyond the normal localized spaces above single element sensors is needed. Distributed sensor systems can be an effective solution for making both proximal and large volume measurements necessary for underwater optical sense and control. Furthermore, spatial light measurements allow optical flux visualizations of water column mass transport, as well as the accurate measurement of chemicals and particles. All of these are key parameters for understanding the subsea environment. Such distributed optical sensor arrays require new systems architectures and new methods of opto-electronic fabrication and construction. In general, light based measurements are the most accessible and cost effective path to creating distributed sensors.

The inventive artificial reef structure allows for the wide distribution of sensors. Natural (biological) systems use distributed sensing and arrays frequently as solutions for sensing or controlling a local environment. Electronic systems similarly use area array architectures also (CSP, BGA's, Flip Chip). Through the use of bio-inspiration, a framework is laid out to base further research on a new style of optical reefs and opto-robotic vehicles (tension-member based drones) for the subsea but the need for new approaches to opto-electronic systems fabrications is needed as well. Using bio-inspiration, radiolarian exoskeletons take on a grid-like 3D pattern creating a cage around the interior and producing strong scaffolding, similar to the string 3D architectural designs. Using the exoskeleton's form and function new styles of sentinel reefs emerge based on open frames with distributed tension members for opto-electronic systems to be embedded, coated or strapped on. These tension frame subsea systems require integrated tension electronic packaging, that is, electronic components, devices and systems that are integral to the tension structural membrane of the sentinel system. In this way the tension member is both mechanical and electronic in nature, similar to PCBs, but shifted to cylindrical geometries which are more hydro-

6

dynamic and deep water pressure friendly. Using flex based packaging substrates as a core building material, and projection optical patterning various components have been fabricated that are planar, tubular and flexible. Three-dimensional digital fabrication may also be used for the production of the rigid structural framework for holding members in tension. The use of three-dimensional printing permits geometrical, shaped bulk objects that can then be paired with flex electronics emerging from printed and flexible electronics.

Integrated electronics and microelectronics are typically designed and fabricated from a planar architecture perspective which then dominates subsequent packaging of such electronic systems. Non planar geometries and wiring are a growing area of electronic design, fabrication and packaging. Novel fibers, printed and flexible sheets allow design of new systems architectures. Additionally, flexible microsystems based on structured, fiber or non-fiber filled PCB laminates and printed electronics permits packaging at the local (micro) substrate scale and at the macro scale with the ability to flex the system across millimeter to centimeter lengths on real everyday systems. Applying such novel electronic packaging strategies (tubular, cylindrical, and flexible sheets) compliments well the undersea environment where planar-boxy and sharp corners prevent survivable pressure and waterproof packaging. Using this microsystems design strategy with modified 2D/3D manufacturing processes a new class of subsea sensors become possible which depart from the classic cylindrical design paradigm. These designs are based on tension frames with electronic micro sensors, components and structures being held in tension as distributed networks of strings, conductors and cables. Hydrodynamic friendly surfaces and subsea sensing systems (both mobile and fixed) emerge from this microelectronics and microsystems packaging approach.

A sentinel reef in an array can form an optical-spectral broadcast system for network communications, underwater drone imaging detectability and tracking in variable environmental conditions. Each cell of the network is an on-board adaptive optical broadcast subsystem that can vary its transmissions spectrally, and spatially using multiple transceivers, and temporally with time-based encodings.

Free space optical wireless communications and networking has a variety of competitive advantages over acoustics including high bandwidth density, security, energy consumption and aesthetics. It uses a highly unregulated part of the spectrum (visible light) for ease of transmissions. Combining acoustics and free space optical communications, can exploit the benefits of complementary path diversity. To ensure maximum robustness to changes in environmental light conditions, adaptive spectral transmissions from the light source are easily deployed using existing low cost multi-color solid-state optoelectronics. Ongoing work focuses on spatial and addressable lighting, waterproofing, light composing, and adaptive low power optoelectronics and integration into underwater test system.

The preceding description contains significant detail regarding the novel aspects of the present invention. It is should not be construed, however, as limiting the scope of the invention but rather as providing illustrations of the preferred embodiments of the invention. Thus, the scope of the invention should be fixed by the claims ultimately drafted, rather than by the examples given.

We claim:

1. An artificial reef structure configured to be moored to the sea floor, comprising:

7

- a. an open frame comprising a plurality of links connected by a plurality of joints;
- b. said open frame including a plurality of rings of said links, with each of said rings lying primarily in a plane, with no two rings lying in the same plane;
- c. said open frame creating a three-dimensional structure;
- d. a connector lying within a central position within said three-dimensional structure;
- e. a plurality of tension members, wherein each of said tension members runs between one of said joints and said connector;
- f. wherein each of said joints, includes,
 - (i) an anchor point attached to one of said tension members,
 - (ii) a first leg, a second leg, and a third leg, with each of said first, second, and third legs including a receiver slidably receiving an end of one of said links,
 - (iii) wherein a first angle between said first leg and said tension member attached to said anchor point is acute,
 - (iv) wherein a second angle between said second leg and said tension member attached to said anchor point is acute,
 - (v) wherein a third angle between said third leg and said tension member attached to said anchor point is acute;

8

- g. said connector being positioned so that every link in said structure is placed in compression by said tension members;
 - h. said tension members lying outside a single plane; and
 - i. a mooring line configured to moor said artificial reef structure above said sea floor and below a sea surface.
- 2.** The artificial reef structure as recited in claim 1, further comprising a float connected to said open frame.
- 3.** The artificial reef structure as recited in claim 1, further comprising:
- (a) a plurality of sheets connected to said open frame; and
 - (b) a light array configured to shine light on a sheet of said plurality of sheets.
- 4.** The artificial reef structure as recited in claim 3, wherein said light array is controlled to produce a moving image upon said sheet.
- 5.** The artificial reef structure as recited in claim 1, further comprising an instrument package configured to monitor water flowing through said open frame.
- 6.** The artificial reef structure as recited in claim 1, further comprising:
- (a) a plurality of sheets connected to said open frame; and
 - (b) a light array configured to shine light on a sheet of said plurality of sheets.

* * * * *